Testicular Dysgenesis Syndrome (TDS) — Comprehensive Disease Characterization

Target disease: Testicular Dysgenesis Syndrome **MONDO ID:** MONDO:0005437 **Category:** Complex / multifactorial developmental syndrome **Report type:** Disease knowledge-base characterization across 15 domains **Investigation:** 5 iterations · 12 confirmed findings · 57 papers reviewed *Evidence base: human clinical/epidemiological studies, rodent model-organism experiments, in-vitro/xenograft studies.*

Summary

Testicular Dysgenesis Syndrome (TDS) is a unifying developmental hypothesis, first articulated by Skakkebaek and colleagues, proposing that four common male reproductive disorders — **cryptorchidism** (undescended testis), **hypospadias**, **impaired spermatogenesis / poor semen quality**, and **testicular germ-cell cancer (TGCT)** — are not independent conditions but four downstream manifestations of a single underlying entity: **disturbed fetal testicular development**, specifically dysfunction of the somatic Sertoli and Leydig cell lineages (PMID: 14671395). These disorders are mutual risk factors, share risk factors (low birth weight, small-for-gestational-age), and co-vary geographically (high in Denmark, low in Finland), supporting a common origin.

The central mechanistic insight established across this investigation is that TDS originates during the **masculinization programming window (MPW)** — roughly gestational weeks 8–14 in humans (embryonic days 15.5–18.5 in the rat) — a critical period during which fetal androgen (testosterone) and INSL3 action program male reproductive-tract development. Deficient androgen action during the MPW, whether caused by **genetic lesions** (NR5A1/SF-1, INSL3/RXFP2, 45,X/46,XY mosaicism, and polygenic TGCT-risk loci such as KITLG, DMRT1, SPRY4, TERT and BAK1) or **anti-androgenic environmental exposures** (phthalate plasticizers, prenatal analgesics/paracetamol) acting on a genetically susceptible background, produces the full TDS phenotype. Reduced **anogenital distance (AGD)** is a validated non-invasive biomarker of this fetal androgen deficit and correlates with the risk and severity of all TDS disorders.

TDS spans a **severity continuum**: from common, mild subfertility at one end, through isolated cryptorchidism/hypospadias, to severe 46,XY gonadal dysgenesis and the most severe form, 45,X/46,XY mixed gonadal dysgenesis, at the other. The **cancer arm** is itself a developmental disease — adult TGCTs arise from a pre-invasive precursor, **germ cell neoplasia in situ (GCNIS)**, derived from arrested fetal gonocytes that retain embryonic pluripotency (OCT4, NANOG, AP-2γ, LIN28, PLAP). Management is **component-specific**: there is no treatment for "TDS" as a unit. Notably, the TGCT component is highly curable (orchiectomy alone cures ~80% of stage I disease), while the fertility and endocrine deficits are largely irreversible. Rising secular trends in TGCT incidence and declining sperm counts worldwide have been interpreted as evidence of a growing environmental contribution to male reproductive health.

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1. Disease Information

Overview. TDS is a constellation / developmental syndrome uniting cryptorchidism, hypospadias, impaired spermatogenesis (poor semen quality), and testicular germ-cell cancer as shared manifestations of disturbed prenatal testicular development (PMID: 14671395; PMID: 16522521). The TDS hypothesis is grounded in epidemiological, clinical, and molecular studies all suggesting that these conditions are interrelated. The most severe form occurs in individuals with 45,X0/46,XY karyotype (mixed gonadal dysgenesis), who present with an intersex phenotype, testicular dysgenesis, undescended testes, hypospadias, severely disrupted spermatogenesis, and high testicular cancer risk (PMID: 16005920).

> *"There is evidence that poor semen quality, testicular cancer, undescended testes and hypospadias are symptoms of one underlying entity, testicular dysgenesis syndrome (TDS), which may be increasingly common due to adverse environmental influences."* — PMID: 14671395

Key identifiers.

Resource	Identifier
MONDO	MONDO:0005437
MeSH	Indexed under gonadal/urogenital dysgenesis concepts; component disorders have dedicated MeSH (Cryptorchidism, Hypospadias, Testicular Neoplasms, Oligospermia/Azoospermia)
ICD-10	Q53 (undescended testicle), Q54 (hypospadias), C62 (malignant neoplasm of testis), N46 (male infertility), Q56 (indeterminate sex)
OMIM	No single OMIM entry for TDS as a syndrome; monogenic contributors include NR5A1 (184757), INSL3 (146738), RXFP2 (606655)

Synonyms / alternative names. Testicular dysgenesis syndrome; TDS; dysgenetic testis syndrome. Component-focused terms: disorders/differences of sex development (DSD) at the severe end.

Information source type. The TDS entity derives predominantly from **aggregated disease-level resources** — epidemiological cohorts, animal models, and molecular/clinical reviews — rather than a single individual-patient registry. Individual-patient data inform the monogenic (DSD) extreme and biomarker studies.

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2. Etiology

Primary causal factors. TDS is **multifactorial**, arising from an interaction of genetic and environmental (particularly endocrine-disrupting) factors that converge on deficient fetal androgen action. "The aetiology of TDS is suspected to be related to genetic and/or environmental factors, including endocrine disrupters" (PMID: 16522521).

Genetic risk factors. - **Monogenic (severe end):** NR5A1/SF-1 (broad 46,XY DSD spectrum — 188 mutations across 238 cases), INSL3 and RXFP2 (cryptorchidism), sex-chromosome mosaicism (45,X/46,XY) (PMID: 31513305; PMID: 41369823; PMID: 16005920). - **Polygenic (cancer arm):** >25 risk SNPs at ~19 loci for TGCT, with per-allele odds ratios among the highest reported for any cancer. Strongest: **KITLG** ($P=2.1\times10^{-48}$), **SPRY4** (1.4×10^{-29}), **DMRT1** (6.7×10^{-25}), **TERT** (1.8×10^{-18}), **BAK1** (2.1×10^{-10}) (PMID: 25598472; PMID: 23640991).

Environmental risk factors. Anti-androgenic endocrine-disrupting chemicals — most prominently **phthalate plasticizers** (DBP, DEHP, DINP) — and **mild analgesics** (paracetamol/acetaminophen, aspirin) taken during pregnancy. Other recognized risk factors include low birth weight and small-for-gestational-age status.

Protective factors. No established genetic protective variants are specifically documented for TDS. By mechanistic logic, avoidance of anti-androgenic exposures during the MPW is protective; the oxidative-stress component of phthalate injury is partly rescued by the antioxidant flavonoid **taxifolin** in a rat model (PMID: 36198370), suggesting antioxidant status may be modifying — but human protective-factor data are lacking.

Gene–environment interactions. The canonical model is that environmental anti-androgens act on a **genetically susceptible** background. A concrete demonstration: maternal mild-analgesic use was dose-dependently associated with cryptorchidism in the Danish but **not** the Finnish arm of a birth cohort (PMID: 21059752). In vitro, inactivating PDE11A variants interact with KITLG risk alleles to elevate familial TGCT risk via cAMP/c-KIT signaling (PMID: 23771924).

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3. Phenotypes

TDS comprises four core phenotypes plus endocrine/biomarker abnormalities:

Phenotype	Type	Onset	Frequency / severity	Suggested HPO
Cryptorchidism (undescended testis)	Physical malformation	Congenital (neonatal)	~1–9% of full-term male infants; uni/ bilateral	HP:0000028
Hypospadias	Physical malformation	Congenital	Variable	HP:0000047
Impaired spermatogenesis / poor semen quality	Laboratory abnormality	Post-puberty/ adulthood	Most common manifestation; severity variable	HP:0000798 (oligospermia)
Testicular germ-cell tumour (seminoma/ nonseminoma)	Neoplasm	Young adult (peak 30–40 y)	Most common malignancy in men 15–45	HP:0010788
Reduced anogenital distance	Physical sign / biomarker	Congenital	Correlates with severity	—
Compensated Leydig-cell failure (low T / high LH)	Laboratory abnormality	Adult	Variable	HP:0008720
Ambiguous genitalia / DSD (severe end)	Physical	Congenital	Rare, severe subset	HP:0000062

Characteristics. Onset of structural components (cryptorchidism, hypospadias) is **congenital**; fertility and cancer components become clinically apparent **post-pubertally / in young adulthood**. Severity is **highly variable** along a continuum. Progression is generally **stable** for the structural malformations (surgically correctable) but the cancer component is progressive if untreated.

Quality-of-life impact. Infertility/subfertility carries psychosocial burden and need for assisted reproduction; TGCT diagnosis in young men has major QoL impact but excellent survival; DSD at the severe end has profound impact on gender, surgical, and psychological care. Per-phenotype validated QoL-instrument (EQ-5D/SF-36) data specific to TDS were not identified.

4. Genetic / Molecular Information

Causal genes (severe/monogenic end). - **NR5A1 (SF-1; OMIM 184757):** essential transcription factor for adrenal/gonadal development and steroidogenesis. 188 mutations across 238 cases span the 46,XY DSD spectrum from severe gonadal dysgenesis to isolated infertility; phenotype includes hypospadias, micropenis, undescended gonads, low AMH/testosterone (PMID: 31513305; PMID: 37409232).

> *"Anti-Müllerian hormone (AMH) and testosterone levels were low, indicating testicular dysgenesis."* — PMID: 37409232

- **INSL3 (OMIM 146738) and RXFP2 (OMIM 606655):** control the trans-abdominal phase of testicular descent. Novel homozygous INSL3 frameshift variants (NM_005543: c.176_182del, p.Ala59GlufsTer66; c.148dupC, p.Arg50ProfsTer16) and recessive RXFP2 variants cause familial bilateral cryptorchidism (PMID: 41369823; PMID: 31167797).

Pathogenic variants — cancer-susceptibility (polygenic). Common low-penetrance variants near **KITLG, SPRY4, DMRT1, TERT-CLPTM1L, BAK1, ATF7IP** predispose to TGCT (predominantly regulatory/intronic SNPs). Replicated in familial and bilateral TGCT: BAK1 rs210138 (OR 1.80), DMRT1 rs755383 (OR 1.67), TERT-CLPTM1L rs4635969 (OR 1.59), KITLG rs2046971 (OR 2.33) (PMID: 21617256). Pediatric and adult germ-cell tumors share overlapping etiology (BAK1, SPRY4, DMRT1, DEPTOR; 18/78 adult loci significant) (PMID: 41941753).

Functional consequences. NR5A1 and INSL3/RXFP2 variants are **loss-of-function**. The RXFP2 D294G missense (mouse ENU model) reduces cell-surface receptor expression via protein misfolding/mistrafficking (PMID: 20963592). TGCT risk SNPs act via altered expression (e.g., KITLG/c-KIT signaling).

Modifier genes. PDE11A inactivating variants modify familial TGCT risk and interact with KITLG (PMID: 23771924). SNP-based susceptibility scores predict pre-GCNIS in androgen-insensitivity syndrome, with KITLG (rs995030) and ATF7IP (rs2900333) prominent (PMID: 29121256).

Epigenetic information. GCNIS/TGCT cells retain an embryonic/pluripotent program (OCT4/POU5F1, NANOG) normally silenced in adult germ cells; vitamin D/VDR signaling downregulates NANOG/OCT4 in vitro (PMID: 23097629). Disease-defining DNA-methylation signatures specific to TDS were not catalogued.

Chromosomal abnormalities. 45,X/46,XY mosaicism (mixed gonadal dysgenesis) is the most severe form; isochromosome 12p is the hallmark somatic marker of invasive TGCT.

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5. Environmental Information

Environmental factors (toxins). Phthalate plasticizers are best-characterized. In rats, in-utero DBP, DEHP, and (more weakly) DINP during the MPW suppress fetal testosterone and cause TDS phenotypes (PMID: 28352662; PMID: 36198370; PMID: 18174951; PMID: 25445723). Combined cadmium + DBP exposure worsens fetal testis defects (PMID: 32497820). **CHEBI suggestions:** dibutyl phthalate (CHEBI:41448), DEHP (CHEBI:17747), cadmium (CHEBI:22977), paracetamol (CHEBI:46195), testosterone (CHEBI:17347).

Important species caveat. Human fetal testis xenografts are **resistant** to phthalate-induced suppression of steroidogenic gene expression, even though they show multinucleated germ-cell induction — demonstrating a **species-specific** sensitivity and a key limitation in extrapolating rat steroidogenic effects to humans (PMID: 22511013).

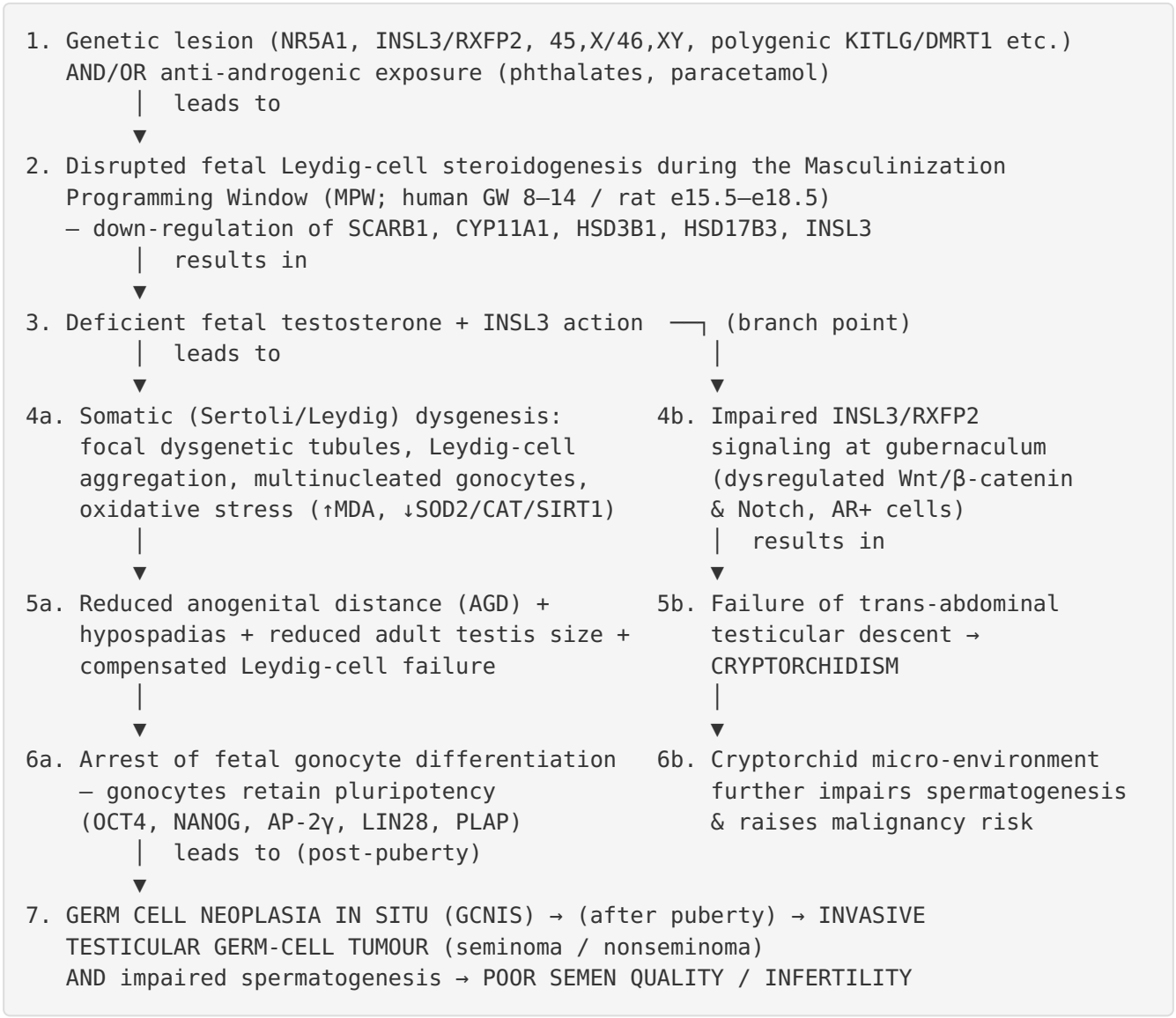
Lifestyle factors. Maternal mild-analgesic (paracetamol, aspirin) use during pregnancy is anti-androgenic and associated with cryptorchidism and shorter infant AGD (PMID: 21059752; PMID: 27609981).

Infectious agents. Not applicable — TDS is a developmental/endocrine disorder with no infectious etiology.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)



Steps 2→3 are **demonstrated** experimentally in rat (MPW-restricted causality of DBP; PMID: 28352662). The gonocyte-arrest → GCNIS → TGCT progression (steps 6a–7) is strongly supported but the malignant-transformation trigger at puberty remains **inferred** (PMID: 18184914).

> *"We show that DBP suppresses fetal testosterone equally during and after the MPW, but only DBP exposure in the MPW causes reduced AGD, focal testicular dysgenesis, and TDS disorders."* — PMID: 28352662

Molecular pathways

- **Steroidogenesis:** SCARB1 → CYP11A1 → HSD3B1 → HSD17B3 cascade producing testosterone; suppressed by phthalates (PMID: 36198370).
- **INSL3–RXFP2 / cAMP** signaling at the gubernaculum, feeding into **Wnt/β-catenin and Notch** pathways and androgen-receptor-positive cell maintenance (PMID: 21147849; PMID: 21403659).
- **KIT/KITLG** receptor-tyrosine-kinase signaling, **telomerase (TERT)**, **DNA-damage repair**, and **microtubule assembly** in TGCT susceptibility (PMID: 25598472).
- **cAMP/PDE11A** interaction with c-KIT in familial predisposition (PMID: 23771924).

Cellular processes & protein dysfunction

Fetal **Leydig-cell** steroidogenic failure and **Sertoli-cell** dysfunction are the upstream somatic lesions; **gonocyte** maturation arrest is the pivotal germ-cell event. GCNIS cells aberrantly express embryonic pluripotency factors (OCT4/POU5F1, NANOG, AP-2γ/TFAP2C, LIN28, PLAP) not seen in normal adult germ cells (PMID: 26306513; PMID: 15982323).

Metabolic / oxidative & tissue-damage mechanisms

Phthalate exposure raises testicular malondialdehyde (lipid peroxidation) and lowers antioxidants (SOD2, CAT, SIRT1, PGC1α); antioxidant rescue (taxifolin) mitigates dysgenesis (PMID: 36198370). Tissue injury manifests as focal dysgenetic tubules, Leydig-cell aggregation, and multinucleated gonocytes.

Suggested ontology terms

- **GO (biological process):** male gonad development (GO:0008584), testosterone biosynthetic process (GO:0061370), Leydig cell differentiation (GO:0033327), Sertoli cell development (GO:0060008), male genitalia development (GO:0030539), response to oxidative stress (GO:0006979).
- **GO (cellular component):** mitochondrion (GO:0005739, steroidogenesis), smooth ER (GO:0005790), nucleus (GO:0005634).

- **CL (cell type):** Leydig cell (CL:0000178), Sertoli cell (CL:0000216), male germ cell / gonocyte (CL:0000015), spermatogonium (CL:0000020).

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7. Anatomical Structures Affected

Organ level. Primary: **testis** (UBERON:0000473), including seminiferous tubules (UBERON:0001343) and the interstitial Leydig compartment. Secondary/associated: **gubernaculum** (descent), **penis/urethra** (UBERON:0000989 / UBERON:0000057; hypospadias), **scrotum** (UBERON:0001300). Body system: **male reproductive / endocrine system** (UBERON:0000079).

Tissue & cell level. Affected cells: **Sertoli cells** (CL:0000216), **Leydig cells** (CL:0000178), **fetal gonocytes / germ cells** (CL:0000015) giving rise to GCNIS. Gubernacular smooth-muscle/mesenchyme and androgen-receptor-positive cells participate in descent (PMID: 21147849).

Subcellular level. Mitochondria and smooth ER of fetal Leydig cells (steroidogenic machinery); nuclear transcription-factor dysfunction (NR5A1, OCT4/NANOG).

Localization / lateralization. Cryptorchidism may be **unilateral or bilateral**; dysgenetic changes are often **focal** within the testis (PMID: 28352662). GCNIS may be unifocal or multifocal.

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8. Temporal Development

Onset. The developmental lesion is **congenital / fetal** (MPW, GW 8–14). Structural components present at birth. Fertility and cancer components become clinically evident **post-pubertally** — TGCT peaks in young adulthood (median diagnosis ~37.7 years; incidence highest ages 30–40) (PMID: 36874603).

Progression. Structural malformations are **stable** and surgically correctable. The cancer arm follows a stepwise **progressive** course: arrested gonocyte → GCNIS (pre-invasive) → invasive TGCT after puberty. Fertility deficits are typically **chronic/lifelong**.

Critical periods. The **MPW (GW 8–14)** is the key window of vulnerability; after this window the same exposure suppresses testosterone but does **not** cause TDS phenotypes (PMID: 28352662). Puberty is the critical window for malignant transformation of GCNIS. The window of opportunity for intervention is **orchidopexy at 6–18 months** to preserve spermatogonia (PMID: 33551366; PMID: 42194159).

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9. Inheritance and Population

Epidemiology. Cryptorchidism affects ~**1–9%** of full-term male infants (2–4% overall; higher in prematurity) (PMID: 42194159; PMID: 20610195). TGCT is the **most common malignancy in men aged 15–45**, with ~500,000 prevalent cases worldwide (2002) and developed-region rates ~6× developing regions (PMID: 17031378). Incidence has risen in successive birth cohorts born from ~1920 onward across 14 countries, with parallel seminoma/nonseminoma trends supporting a common fetal rate-limiting step.

> *"incidence has uniformly increased in successive generations born from around 1920 until very recently"* — PMID: 17031378

Inheritance patterns. Mixed: the monogenic DSD end can be **autosomal dominant** (many NR5A1), **autosomal recessive** (INSL3/RXFP2 familial cryptorchidism), or **sex-chromosome mosaic** (45,X/46,XY). Common forms are **multifactorial / polygenic** — most isolated cryptorchidism is multifactorial (PMID: 21403659), and TGCT susceptibility is polygenic with high per-allele ORs (PMID: 25598472).

Penetrance / expressivity. NR5A1 shows highly **variable expressivity** (same gene → severe gonadal dysgenesis to isolated infertility) and incomplete penetrance (PMID: 31513305). Consanguinity contributes to recessive INSL3/RXFP2 cryptorchidism (PMID: 31167797).

Population demographics. Predominantly affects **Western/Caucasian** populations; historically highest national incidence in **Denmark/Norway**, lowest in **Finland** — mirroring the north-European gradient in cryptorchidism, hypospadias, and semen quality (PMID: 17031378; PMID: 16005920). By definition the disorder affects genetic/phenotypic males.

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10. Diagnostics

Hormonal / laboratory biomarkers. - **Sertoli-cell markers:** inhibin B and anti-Müllerian hormone (AMH). Low inhibin B/FSH and AMH/FSH ratios and high FSH predict impaired germ cells per tubule in bilateral cryptorchid infants (ROC predictors, p=0.004–0.019) (PMID: 34823367). AMH correlates with testicular volume and sperm concentration in men with maldescended testes (PMID: 18423454).

> *"Anti-Müllerian hormone and inhibin B are essential biomarkers of Sertoli cell function. Also, INSL3 arises as an additional marker of Leydig cell dysfunction."* — PMID: 23549308

- **Leydig-cell markers:** testosterone, LH, INSL3.
- **Semen analysis:** sperm concentration/count per WHO criteria.
- **Anogenital distance (AGD):** non-invasive biomarker of fetal androgen action (PMID: 26846869; PMID: 23569219).

Imaging. Scrotal/inguinal ultrasound; MRI for non-palpable testis.

Genetic testing. Karyotype (detects 45,X/46,XY), chromosomal microarray, and DSD gene panels (NR5A1, INSL3, RXFP2, and other DSD genes); WES/targeted NGS resolves 46,XY DSD etiology (PMID: 34338568).

Histopathology / GCNIS detection. Testicular biopsy with immunohistochemistry for pluripotency markers **OCT4/POU5F1, PLAP, AP-2γ, LIN28**; these can also detect GCNIS in semen samples (PMID: 26306513). Pre-GCNIS shows centrally located OCT3/4+ gonocytes without full GCNIS morphology (PMID: 40767035).

Differential diagnosis. Klinefelter syndrome, Noonan syndrome (Sertoli-cell-specific primary testicular insufficiency; PMID: 30325180), androgen insensitivity syndrome, persistent Müllerian duct syndrome (PMID: 40767035), and isolated (non-syndromic) component disorders.

11. Outcome / Prognosis

TGCT component — excellent. "Orchiectomy alone cure 80% of stage I" disease (PMID: 31582176). Adjuvant treatments reduce relapse but confer **no overall-survival benefit** with substantial toxicity, so surveillance is now favored. TGCT is among the most curable solid cancers.

> *"Adjuvant treatments decrease the relapse rate after orchiectomy with substantial toxicities without any benefit on overall survival. Recent guidelines accorded utmost importance on SV rather than adjuvants strategies."* — PMID: 31582176

Cryptorchidism component. Even after orchidopexy, long-term outcomes can be problematic: impaired fertility (~33% unilateral, ~66% bilateral) and a **5–10× increased malignancy risk** (PMID: 20610195). Formerly unilateral cryptorchid men have only slightly (non-significantly) reduced paternity rates (PMID: 33551366).

Fertility / endocrine components. Subfertility and compensated Leydig-cell failure are **largely irreversible**, managed by assisted reproduction and androgen replacement.

Prognostic factors. Low inhibin B/AMH, high FSH, reduced AGD, bilaterality, and Leydig-cell function (predicts sperm-retrieval success in Klinefelter: PMID: 26235799).

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12. Treatment

Management is component-specific — there is no unitary "TDS" therapy.

Component	Treatment	NCIT suggestion
Cryptorchidism	**Orchidopexy** at 6–18 mo (ideally <12 mo) to preserve spermatogonia and reduce torsion/subfertility/malignancy risk (PMID: 42194159; PMID: 33551366). Hormonal therapy (hCG/GnRH) has limited efficacy, not recommended as primary treatment (PMID: 23748056)	Orchiopexy (C51919)
TGCT	**Radical inguinal orchiectomy** first-line; stage I options: active surveillance, single-agent carboplatin (seminoma) or BEP (bleomycin-etoposide-cisplatin)/RPLND (nonseminoma) (PMID: 31582176)	Orchiectomy (C51576), Cisplatin (C376)
Hypospadias	Surgical urethroplasty	Urethroplasty
Infertility / poor semen quality	Assisted reproduction (ICSI/IVF), testicular sperm extraction (TESE/mTESE)	ICSI, IVF
Androgen insufficiency	Testosterone replacement	Testosterone therapy (C548)

> *"Early referral, ideally before 12 months of age, is essential for timely orchidopexy, which aims to position the testes within the scrotum to reduce risks of torsion, trauma, subfertility, and malignancy."* — PMID: 42194159

Pharmacogenomics / experimental. Taxifolin (antioxidant) rescues phthalate-induced dysgenesis in rats — a mechanistic lead, not a human therapy (PMID: 36198370). Vitamin D/VDR downregulates pluripotency factors in TGCT cells in vitro but did not reduce tumor growth in xenografts (PMID: 23097629).

13. Prevention

Primary prevention. Reduce maternal exposure to anti-androgenic endocrine disruptors (phthalates) and avoid non-essential **mild analgesics (paracetamol)** during pregnancy, especially during the MPW (GW 8–14) (PMID: 21059752; PMID: 27609981). Environmental/regulatory interventions limiting phthalate exposure are logical public-health measures.

Secondary prevention. Early detection and **timely orchidopexy** of cryptorchidism; inclusion of testicular-cancer surveillance in the management of infertile men with poor semen quality (PMID: 16522521). GCNIS detection via biopsy/semen immunocytochemistry in high-risk individuals (PMID: 26306513).

Tertiary prevention. Surveillance of the contralateral testis and GCNIS monitoring; fertility preservation counseling.

Genetic counseling. Relevant for monogenic DSD (NR5A1, INSL3/RXFP2) and familial/bilateral TGCT. Prophylactic gonadectomy decisions in AIS should be individualized given low observed malignant progression (PMID: 29121256).

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14. Other Species / Natural Disease

Taxonomy / orthologs. Rat (*Rattus norvegicus*, NCBI:txid10116) and mouse (*Mus musculus*, NCBI:txid10090) are the principal experimental species. Orthologous genes: *Insl3*, *Rxfp2*, *Nr5a1*, *Kitlg*, *Dmrt1*.

Comparative biology — key caveat. Rats show robust phthalate-induced steroidogenic suppression, mice show only multinucleated germ-cell induction, and **human fetal testis xenografts resist steroidogenic suppression** — a critical species difference (PMID: 22511013; PMID: 32779695). This limits direct extrapolation of rodent androgen-suppression mechanisms to humans, though the MPW concept and AGD biomarker translate.

Natural disease / zoonotic. No naturally occurring companion-animal or zoonotic form of TDS was identified; this is not a transmissible disease.

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15. Model Organisms

Rat (principal model). In-utero phthalate (DBP/DEHP/DINP) exposure during the MPW is the gold-standard **induced model**, recapitulating reduced AGD, cryptorchidism, hypospadias, reduced adult testis size, compensated Leydig-cell failure, focal dysgenesis, Leydig-cell aggregation, and multinucleated gonocytes (PMID: 28352662; PMID: 18174951; PMID: 33086118). **High phenotypic recapitulation** of the structural TDS components; focal dysgenetic changes resemble human pre-invasive lesions.

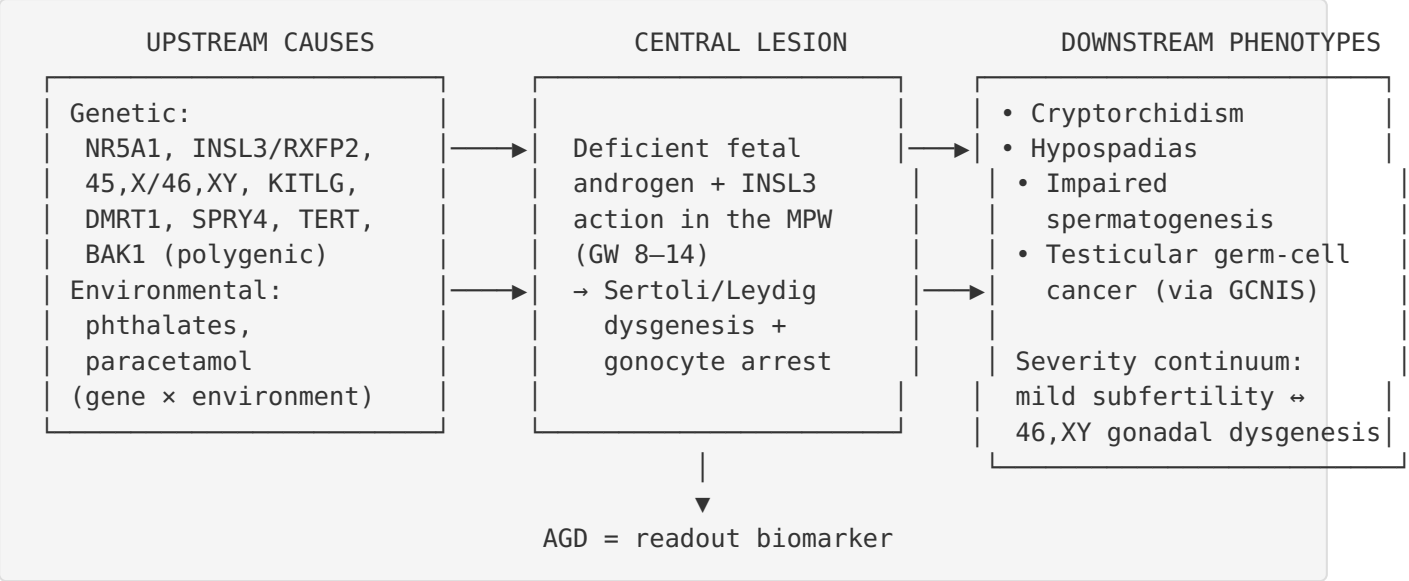
Mouse (genetic models). *Insl3* and *Rxfp2* knockouts cause cryptorchidism; the ENU **Rxfp2 D294G** missense model impairs descent with Sertoli-cell-only tubules (PMID: 20963592); conditional shRNA RXFP2 suppression reveals Wnt/β-catenin and Notch involvement (PMID: 21147849).

Cell lines (in vitro). NTERA-2 (embryonal carcinoma) and TCam-2 (seminoma) for TGCT pluripotency/differentiation and KITLG/cAMP studies (PMID: 23097629; PMID: 23771924).

Model limitations. Rodents do **not** fully replicate the human germ-cell-cancer arm (no spontaneous GCNIS→TGCT progression), and rodent steroidogenic sensitivity to phthalates exceeds that of humans — a major translational caveat (PMID: 22511013).

Mechanistic Model / Interpretation

TDS is best understood as a **single fetal-origin lesion with four clinical faces**. The unifying node is **insufficient androgen (and INSL3) action in the masculinization programming window**. Everything upstream (genetic lesions, endocrine disruptors) converges on this node; everything downstream (the four disorders) diverges from it:



Two mechanistic subtleties matter for the knowledge base. First, the **cancer arm is developmental**: GCNIS is an arrested, pluripotent fetal gonocyte that only becomes malignant after puberty — explaining both the young-adult incidence peak and the shared pluripotency biomarkers (OCT4/NANOG/AP-2γ/LIN28/PLAP). Second, the **human–rodent discordance** in

phthalate steroidogenic sensitivity means the rat model proves *that* MPW androgen suppression causes TDS, but human causation by specific chemicals rests on epidemiology and biomarker (AGD) association rather than direct experimental proof.

Evidence Base

PMID	Role	Key contribution
14671395	Foundational	Original TDS hypothesis uniting four disorders
16005920	Supporting	Severe form (45,X/46,XY); geographic co-variation
28352662	Mechanistic (strong)	MPW-restricted causality of DBP-induced TDS
32779695	Mechanistic	MPW androgen action; AGD biomarker; human-rodent differences
36198370	Mechanistic	Steroidogenic gene suppression + oxidative stress; taxifolin rescue
18174951	Dose-response	Fetal endpoints most sensitive to DBP
22511013	Challenging/caveat	Human fetal testis resists phthalate steroidogenic suppression
25598472	Genetic	>25 TGCT risk SNPs at 19 loci; 5 mechanistic categories
23640991	Genetic	Gene-level P-values (KITLG, DMRT1, SPRY4, TERT, BAK1)
21617256	Genetic	Replication in familial/bilateral TGCT
31513305	Genetic (DSD)	188 NR5A1 mutations; phenotypic spectrum
41369823	Genetic	Novel INSL3 LoF variants → cryptorchidism
21403659	Mechanistic	INSL3/testosterone control of descent; multifactorial
21059752	Gene×environment	Analgesics → cryptorchidism (Danish vs Finnish)
27609981	Human biomarker	Prenatal paracetamol → shorter AGD
18184914	Mechanistic (cancer)	CIS/GCNIS from arrested fetal germ cells
26306513	Diagnostic	Pluripotency biomarkers for GCNIS
15982323	Mechanistic	NANOG in fetal gonocytes, CIS, GCT
31582176	Prognosis/treatment	Orchiectomy cures 80% stage I; surveillance favored
42194159	Treatment/epi	Cryptorchidism 1–9%; orchidopexy <12 mo
34823367	Diagnostic	Inhibin B/FSH, AMH/FSH predict germ-cell deficiency
23549308	Diagnostic	AMH, inhibin B, INSL3 as Sertoli/Leydig markers

PMID	Role	Key contribution
28981654	Epidemiology	Sperm-count decline meta-regression
36377604	Epidemiology	Global extension of sperm-count decline
17031378	Epidemiology	Geographic/birth-cohort trends in TGCT
36874603	Epidemiology	Median TGCT diagnosis age 37.7 y

Limitations and Knowledge Gaps

1. **TDS is a hypothesis, not a discrete molecular diagnosis.** It lacks a single OMIM entry, defined diagnostic criteria, or a unitary genetic test; the four components are also studied as independent disorders. The unifying construct rests on epidemiological co-occurrence and a shared fetal-androgen mechanism.
2. **Human causation by specific chemicals is inferential.** The strongest experimental evidence (MPW-restricted phthalate causality) is from rats; human fetal testis resists phthalate steroidogenic suppression (PMID: 22511013). Human evidence is association-based (AGD, cohort studies).
3. **The malignant-transformation trigger** converting quiescent GCNIS to invasive TGCT after puberty is not fully defined.
4. **Genetic architecture is incompletely explained.** Known INSL3/RXFP2 mutations account for only a minority of cryptorchidism; TGCT heritability is polygenic and partly "missing."
5. **Epigenetic and single-cell/spatial profiling** specific to human TDS tissues were not catalogued and represent a gap.
6. **Quality-of-life data** using standardized instruments (EQ-5D/SF-36) specific to TDS were not identified.
7. **No protective genetic variants** have been established.

Proposed Follow-up Experiments / Actions

1. **Human fetal testis organoids / xenografts** to define the molecular basis of the human-specific resistance to phthalate steroidogenic suppression, reconciling rodent and human mechanisms.
2. **Single-cell and spatial transcriptomics** of dysgenetic human fetal/perinatal testis to map Sertoli/Leydig/gonocyte trajectories and the gonocyte-arrest signature at cell resolution.
3. **Prospective AGD-anchored birth cohorts** linking MPW-window chemical/analgesic exposure, AGD, germline genotype (KITLG/DMRT1/NR5A1 polygenic score), and adult reproductive outcomes to formally test gene×environment interaction.
4. **GCNIS progression biomarkers:** define the pubertal signals (and candidate circulating/semen miRNAs) that drive GCNIS→TGCT, enabling earlier risk stratification.
5. **Antioxidant intervention proof-of-concept:** extend the taxifolin rescue finding (PMID: 36198370) toward evaluating maternal antioxidant status as a modifier of endocrine-disruptor injury.
6. **Harmonized diagnostic panel:** standardize combined inhibin B/AMH/FSH/INSL3 + AGD + pluripotency-marker IHC into a validated TDS-risk stratification tool.
7. **Knowledge-base curation:** map the suggested ontology terms (HPO, GO, CL, UBERON, CHEBI, NCIT) to each phenotype/mechanism/cell type for structured annotation.

Report generated from a 5-iteration autonomous investigation: 12 confirmed findings, 57 papers reviewed. Evidence types span human clinical/epidemiological, model organism (rat/mouse), in vitro, and computational/GWAS sources, distinguished throughout.
